A Proof of the Sum of Cubes Formula

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Consider the sum

$$\mathscr{P}(n) := \sum_{i=0}^{n} i^3$$

As a matter of mathematical curiosity, our aim is to find and prove the validity of a closed formula for $\mathscr{P}(n)$, for all $n \in \mathbb{N}$ (following the convention of Lean and type theory, we consider 0 to be a member thereof). A plausible place to start is simple, direct computation of the first few terms. We see by mental arithmetic that the first 10 terms are

$$0, 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000$$

and so, we have that the first ten values of $\mathscr{P}(n)$ are

$$0, 1, 9, 36, 100, 225, 441, 784, 1296, 2025, 3025$$

These are all, evidently, square numbers. So, we propose that there exists some sequence $\mathscr{S}(n)$ such that $\mathscr{P}(n) = \mathscr{S}^2(n)$. The next natural step is to try to see if there exists some (ideally formulaic) pattern between the values of $\mathscr{S}$, which might lead to development of our desired closed formula for $\mathscr{P}(n)$. By inspection, we see that the first 10 values of $\mathscr{S}(n)$ are

$$0, 1, 3, 6, 10, 15, 21, 28, 36, 45, 55$$

Again by inspection, we see that the $n + 1^{\text{th}}$ term is itself the sum of the previous n terms. Though this is a problem that already has a well-known solution, the fact that this exercise will eventually require the entirety of the nontrivial parts of the proof guides us to a derivation thereof; that is, of the formula for the sum of all integers up to n (including 0). Consider the sum of the sequence of the integers up to 10:

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$$

where, since it is the additive identity, 0 may be excluded from consideration. Now, notice that, if we start from the "ends" of the sequence (ignoring 0), and compute the sum of the sequence as the sum of the pairs of the distinct elements "equidistant" from either end (i.e. $1 + 10, 2 + 9, ...$) we see that they all have the same sum. So, for this particular case, we have that the sum of the integers up to 10 is $\left(\frac{10}{2}\right)(11) = 55$. It is easy to see, at least for such small n as would permit mental computation, that the same logic holds. Therefore, we propose the following

$$\mathscr{S}(n) = \sum_{i=0}^{n} i = \frac{n(n+1)}{2}$$

– a statement which we now prove.

The Sum of the Naturals up to n

Proof. Let us define, for all $n \in \mathbb{N}$, P_n to be the statement that the sum of the naturals up to n is, as we propose, $\frac{n(n+1)}{2}$. Thus, our aim is to show that P_n holds for all n. We start with the base case, wherein $n = 0$. Clearly,

$$\sum_{i=0}^{0} i = 0$$

and so P_n holds in the base case.

We now propose our inductive hypothesis; that, for all $k \in \mathbb{N}$, P_k holds, and subsequently prove that, given the validity of P_k, P_{k+1} must also be true. By definition of $\mathscr{S}(n)$, we have that

$$\mathscr{S}(k+1) = \sum_{i=0}^{n} i + (k+1)$$

which, by our assumption of the truth of P_k, becomes

$$\frac{k(k+1)}{2} + (k+1) = \left(\frac{(k+1)(k+1+1)}{2}\right)$$

As such, we have that P_{k+1} is true.

Therefore, by the Principle of Mathematical Induction, we have that, for all $n \in \mathbb{N}$,

$$\mathscr{S}(n) = \frac{n(n+1)}{2}$$

as desired. $\square$

The Sum of the Cubes of the Naturals up to n

We now return to our previous proposition that $\mathscr{P}(n) = \mathscr{S}^2(n)$, and formulate a proof in a similar style to that just completed.

Proof. Let us define, for all $n \in \mathbb{N}$, the statement P_n to be that $\mathscr{P}(n) = \mathscr{S}^2(n)$. Our goal is, then, to prove that P_n holds for all $n \in \mathbb{N}$.

As is customary for proofs by induction, we begin with the base case. Clearly, by the definition of $\mathscr{P}(n)$, we see that

$$\mathscr{P}(0) = \sum_{i=0}^{0} i = 0 = \left(\frac{0(0+1)}{2}\right)^2 = \mathscr{S}^2(0)$$

– thus, we have proven the base case.

We now assume, as our inductive hypothesis, that, for all $k \in \mathbb{N}$, P_k holds true, and use this assumption as a basis from which to prove that P_{k+1} must then always hold. By definition, we have that

$$\mathscr{P}(k+1) = \sum_{i=0}^{k} i^3 + (i+1)^3$$

Then, given our assumption of the truth of P_k, this becomes

$$\left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 = \frac{1}{4}(k+1)^2\left(k^2 + 4(k+1)\right)$$

$$= \left(\frac{(k+1)(k+2)}{2}\right)^2 = \left(\mathscr{S}(k+1)\right)^2$$

Therefore, by the Principle of Mathematical Induction, we have shown that, for all $n \in \mathbb{N}$,

$$\mathscr{P}(n) = \mathscr{S}^2(n)$$

as desired. Furthermore, as a final note, we have also arrived precisely at our original goal – namely, to find an explicit formula for $\mathscr{P}(n)$. Specifically, substituting our known formula for $\mathscr{S}(n)$, we have, for all natural n, the elegant expression

$$\mathscr{P}(n) = \left(\frac{n(n+1)}{2}\right)^2$$

$\square$